

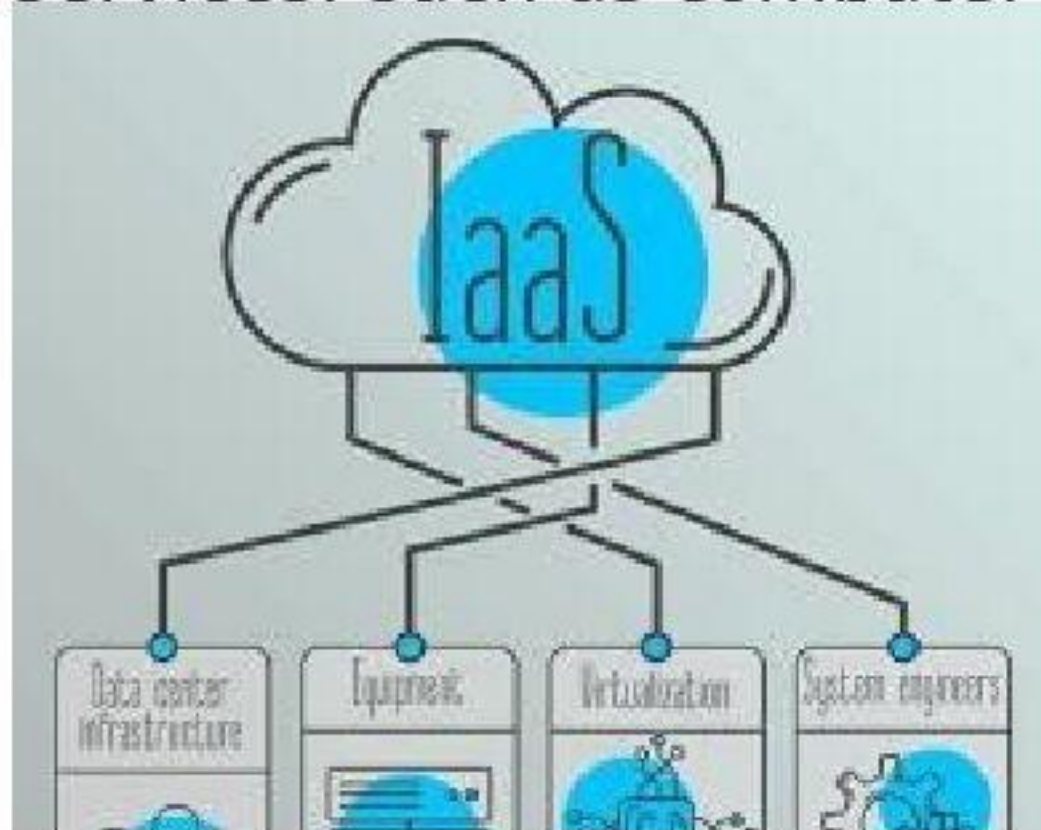
CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

Microsoft Cloud Services is a suite of cloud computing services from Microsoft that provide a wide range of capabilities, including:

- Infrastructure as a Service (IaaS): Azure offers a variety of IaaS services, such as compute, storage, and networking. These services allow you to build and deploy cloud-based applications



Microsoft Azure IaaS

- Platform as a Service (PaaS): Azure offers a variety of PaaS services, such as app development, data management, and business intelligence. These services can be used to develop and deploy cloud-based applications without having to manage the underlying infrastructure.



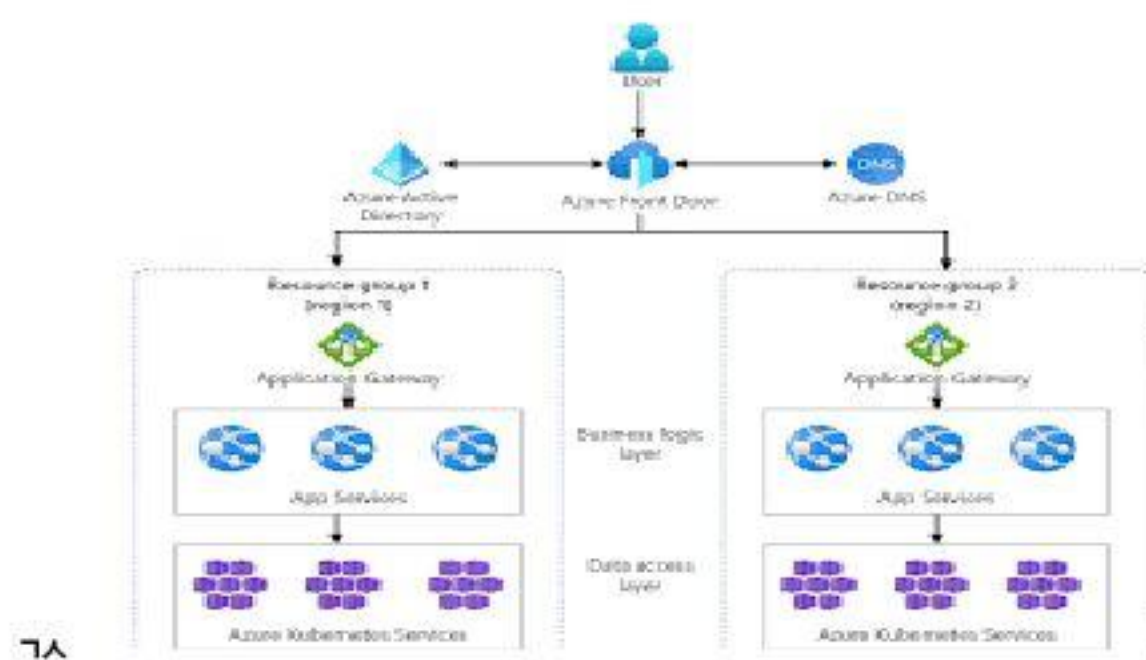
Microsoft Azure PaaS

- Software as a Service (SaaS): Azure offers a variety of SaaS applications, such as Office 365, Dynamics 365, and Azure Sentinel. These applications can be accessed from any device with an internet connection.

CLOUD COMPUTING-II

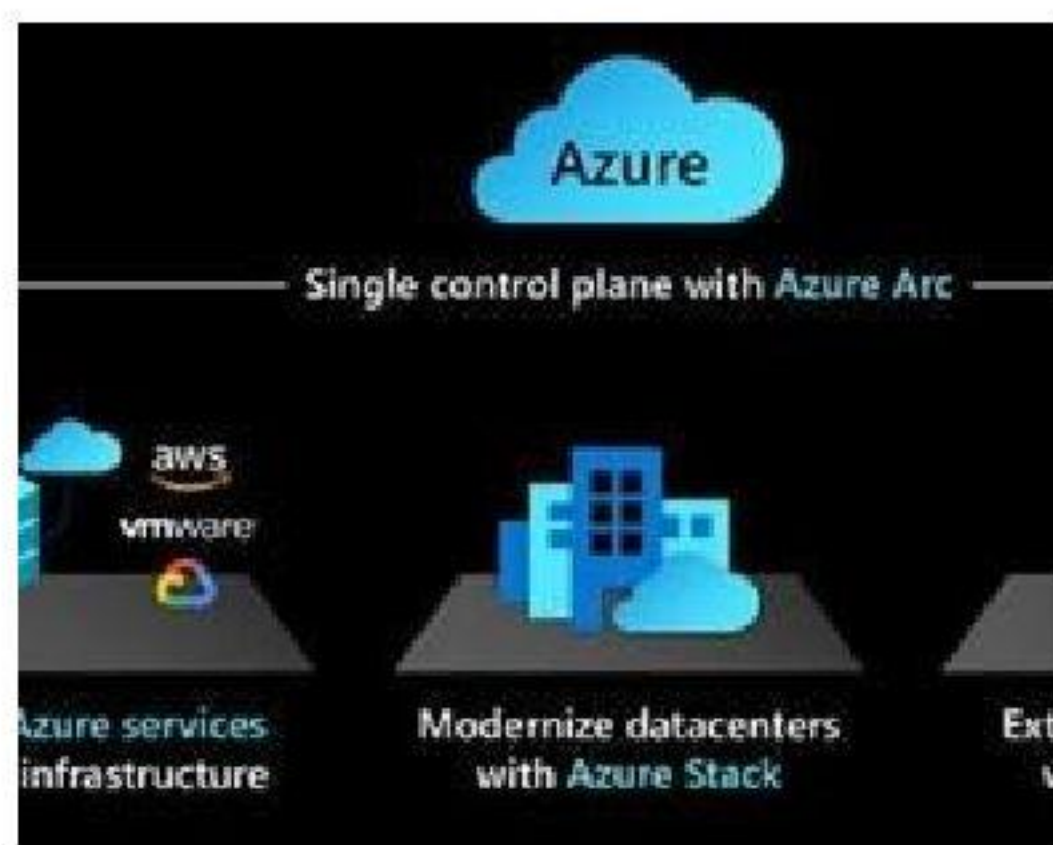
UNIT-III

Microsoft Cloud Services



Microsoft Azure SaaS

- 7A Hybrid and multi-cloud: Microsoft Azure supports hybrid and multi-cloud deployments, which allow you to use a combination of on-premises and cloud-based resources. This can be helpful if you have existing on-premises infrastructure that you want to keep, or if you want to use a combination of cloud providers.



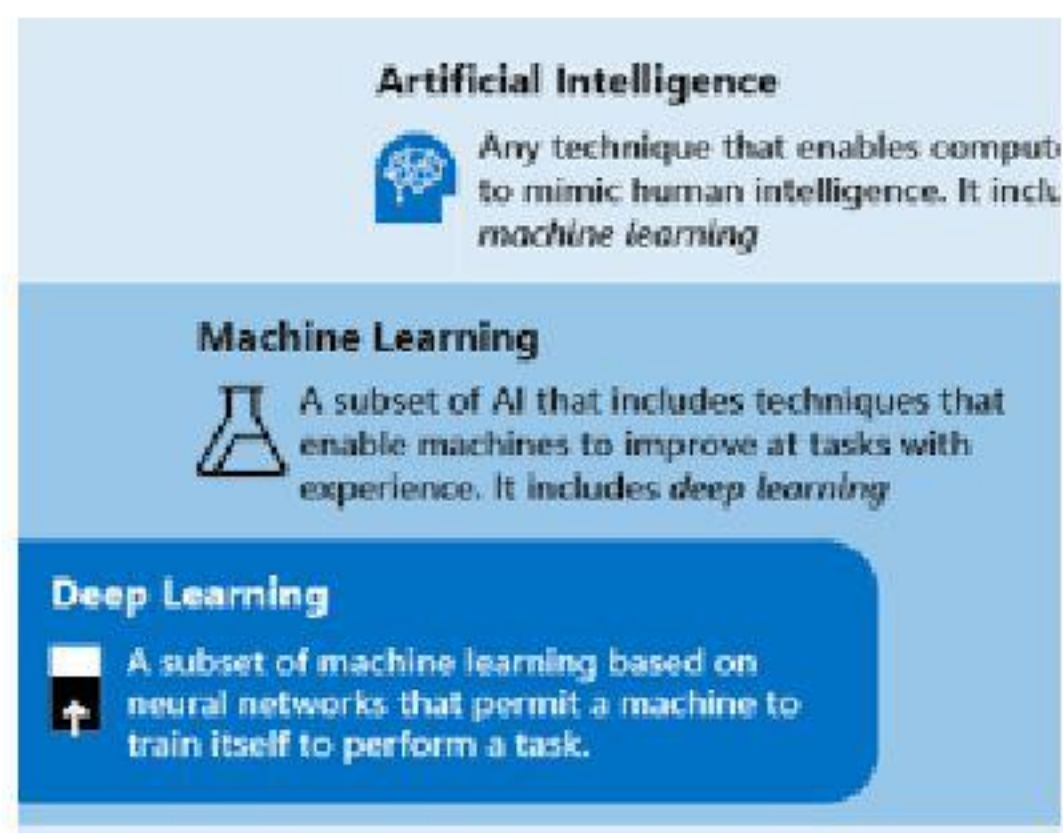
Microsoft Azure Hybrid and Multi Cloud

- 7A Artificial intelligence and machine learning: Azure offers a variety of AI and machine learning services, such as Azure Cognitive Services, Azure Machine Learning, and Azure Bot Service. These services can be used to develop intelligent applications that can learn from data and make predictions.

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

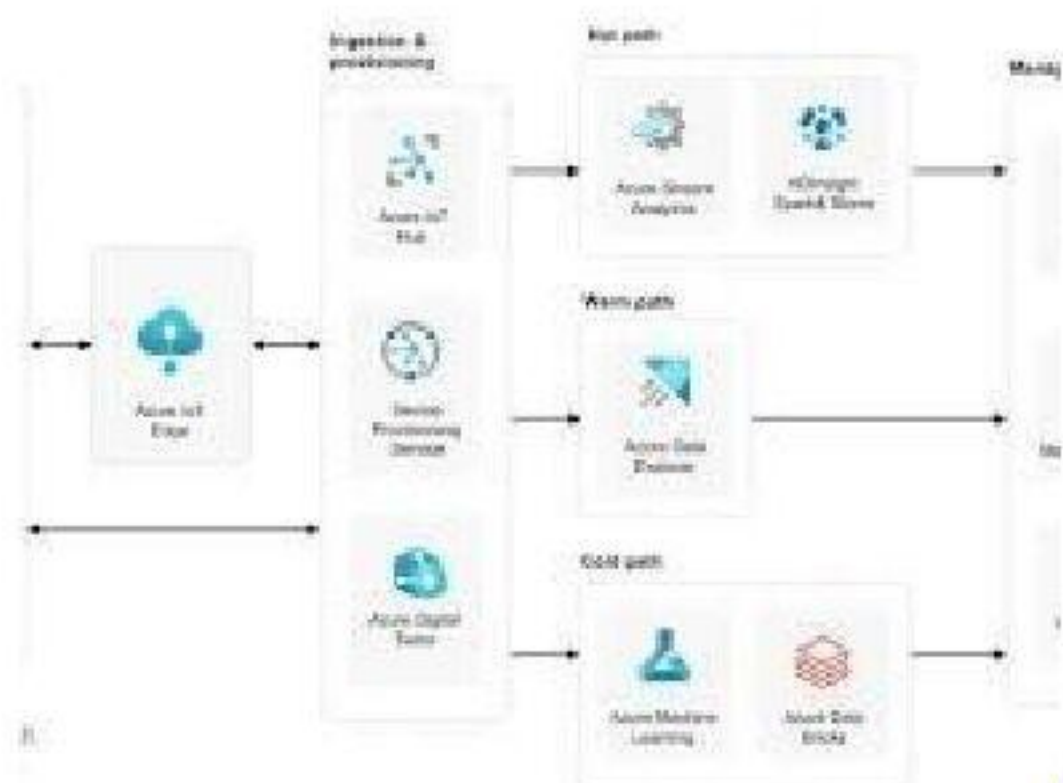


[Opens in a new window](#)

learn.microsoft.com

Microsoft Azure Artificial Intelligence and Machine Learning

- Internet of Things (IoT): Azure offers a variety of IoT services, such as Azure IoT Hub, Azure IoT Edge, and Azure IoT Central. These services can be used to connect, monitor, and manage IoT devices.



[Opens in a new window](#)

learn.microsoft.com

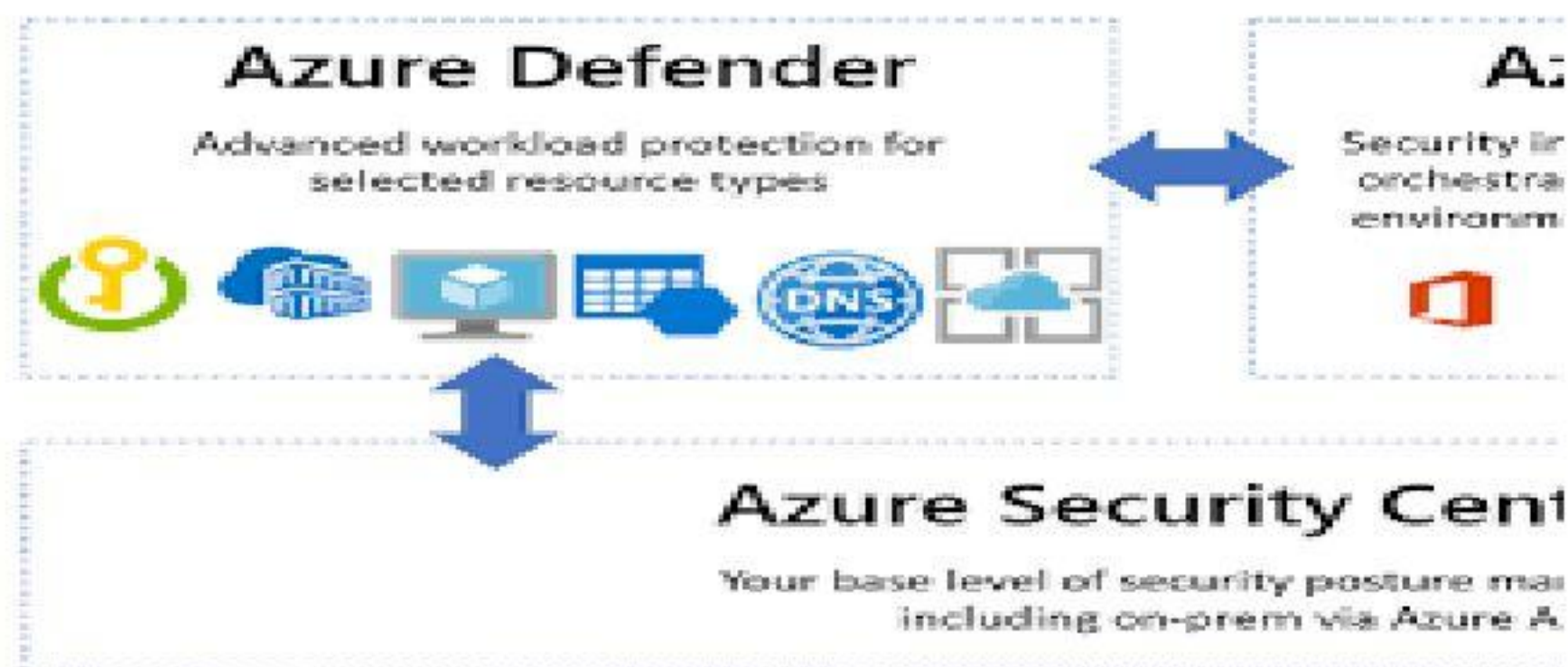
Microsoft Azure IoT

- Security: Azure offers a variety of security services, such as Azure Security Center, Azure Defender, and Azure Sentinel. These services can be used to protect your cloud-based resources from cyberattacks.

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services



Microsoft Azure Security

Microsoft Cloud Services can be a valuable asset for businesses of all sizes. They can help you to:

- Reduce costs: By using cloud-based resources, you can avoid the costs of purchasing and maintaining on-premises infrastructure.
- Improve agility: Cloud-based resources can be provisioned and scaled quickly and easily, which can help you to respond to changes in demand.
- Increase productivity: Cloud-based applications can be accessed from anywhere with an internet connection, which can help your employees to be more productive.
- Enhance security: Cloud-based resources are often more secure than on-premises resources, as they are managed by experts who are constantly working to protect them from cyberattacks.
- Empower innovation: Cloud-based services can provide you with the tools and resources you need to innovate and develop new products and services.

If you are considering using Microsoft Cloud Services, I recommend that you contact a Microsoft partner to discuss your specific needs. They can help you to choose the right services for your business and can provide you with the support you need to get started.

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

Understanding Microsoft Azure platform

Microsoft Azure, formerly known as Windows Azure, is a cloud computing platform and web services provided by Microsoft. It provides software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS) with over 200 cloud services.

Key Features of Microsoft Azure:

- ↯ Scalability: Azure can be scaled up or down based on demand, making it a cost-effective solution for businesses of all sizes.
- ↯ Reliability: Azure has a global network of data centers, which ensures high availability and low latency for users around the world.
- ↯ Security: Azure offers a variety of security features, such as encryption, access control, and data loss prevention.
- ↯ Compliance: Azure is compliant with a wide range of industry standards, including HIPAA, PCI DSS, and SOC 2.
- ↯ Open source: Azure supports a wide range of open source technologies, such as Linux, Docker, and Kubernetes.

Services Offered by Microsoft Azure:

- ↯ Compute: Azure offers a variety of compute services, such as virtual machines, containers, and serverless computing. These services can be used to run a wide range of applications, from web applications to batch processing jobs.
- ↯ Storage: Azure offers a variety of storage services, such as blobs, queues, and tables. These services can be used to store a wide range of data, from unstructured data to structured data.
- ↯ Networking: Azure offers a variety of networking services, such as virtual networks, firewalls, and load balancers. These services can be used to connect and secure cloud-based applications.
- ↯ Data management: Azure offers a variety of data management services, such as databases, analytics, and machine learning. These services can be used to collect, store, analyze, and act on data.
- ↯ Application development: Azure offers a variety of application development services, such as DevOps, API management, and bot services. These services can be used to develop, deploy, and manage cloud-based applications.

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

Benefits of Using Microsoft Azure:

- Reduced costs: Azure can help businesses reduce costs by eliminating the need to purchase and maintain on-premises hardware and software.
- Increased agility: Azure can help businesses increase agility by allowing them to quickly provision and scale cloud resources.
- Improved productivity: Azure can help businesses improve productivity by providing employees with access to cloud-based applications and data from anywhere in the world.
- Enhanced security: Azure can help businesses enhance security by providing a variety of security features, such as encryption, access control, and data loss prevention.
- Empowered innovation: Azure can help businesses empower innovation by providing them with the tools and resources they need to develop new products and services.

Use Cases of Microsoft Azure:

- Web applications: Azure can be used to develop and deploy web applications that are scalable, reliable, and secure.
- Mobile applications: Azure can be used to develop and deploy mobile applications that are scalable, reliable, and secure.
- Internet of Things (IoT): Azure can be used to connect, manage, and analyze data from IoT devices.
- Big data: Azure can be used to collect, store, analyze, and visualize big data.
- Machine learning: Azure can be used to develop and deploy machine learning models.

Overall, Microsoft Azure is a powerful and versatile cloud computing platform that can be used to address a wide range of business needs.

Core Azure Services

Microsoft Azure offers a variety of core services that can be used to build and deploy cloud-based applications and infrastructure. These services can be broadly categorized into four groups: compute, networking, storage, and databases.

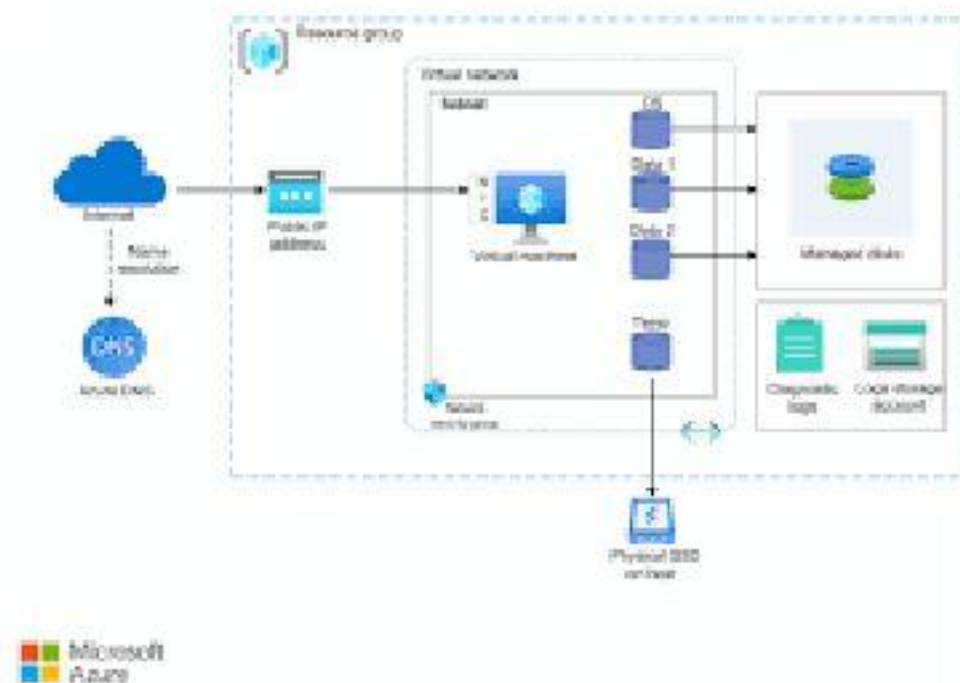
Compute

- Virtual Machines (VMs): VMs are the basic building blocks of Azure compute. They allow you to run any software on a virtual server that is hosted in the cloud.

CLOUD COMPUTING-II

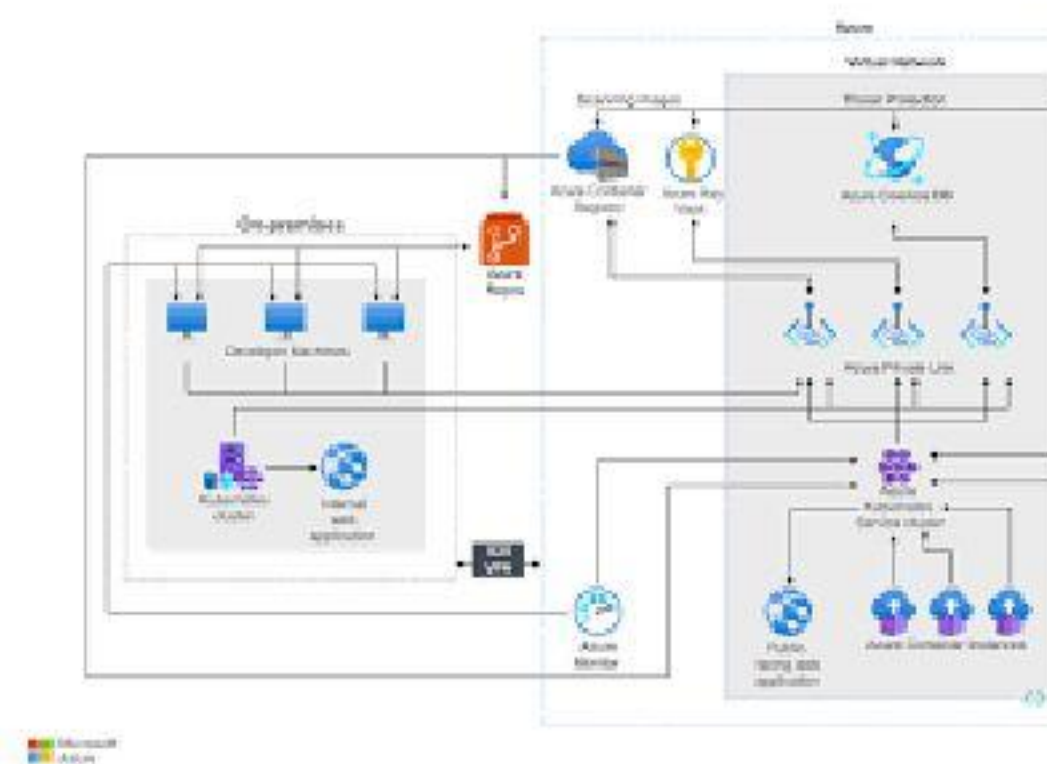
UNIT-III

Microsoft Cloud Services



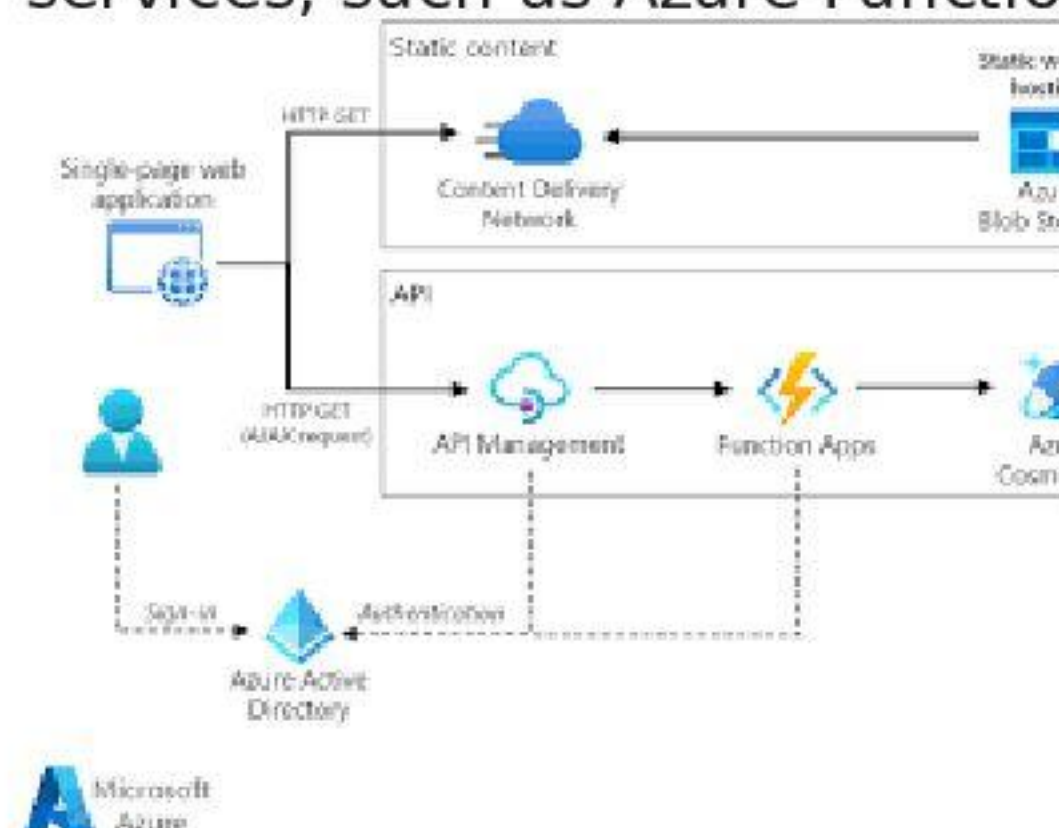
Microsoft Azure Virtual Machines

- Containers: Containers are a lightweight, portable way to package and deploy applications. Azure offers a variety of container services, such as Azure Container Instances and Azure Kubernetes Service (AKS).



Microsoft Azure Containers

- Serverless Computing: Serverless computing allows you to run code without having to provision or manage servers. Azure offers a variety of serverless computing services, such as Azure Functions and Azure Logic Apps.



CLOUD COMPUTING-II

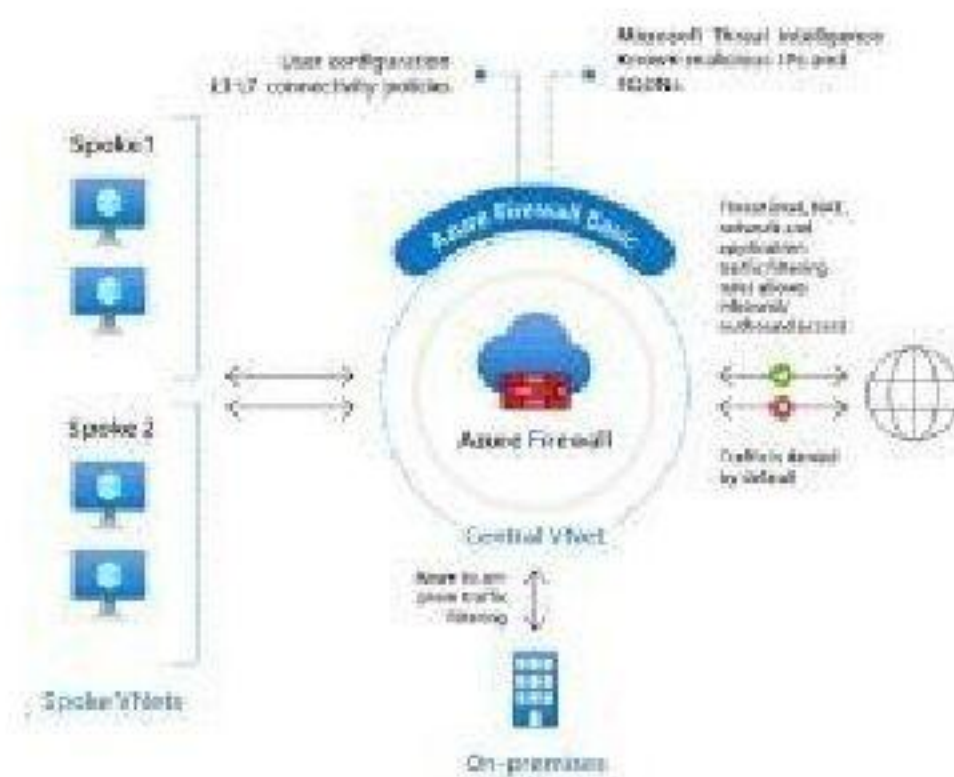
UNIT-III

Microsoft Cloud Services

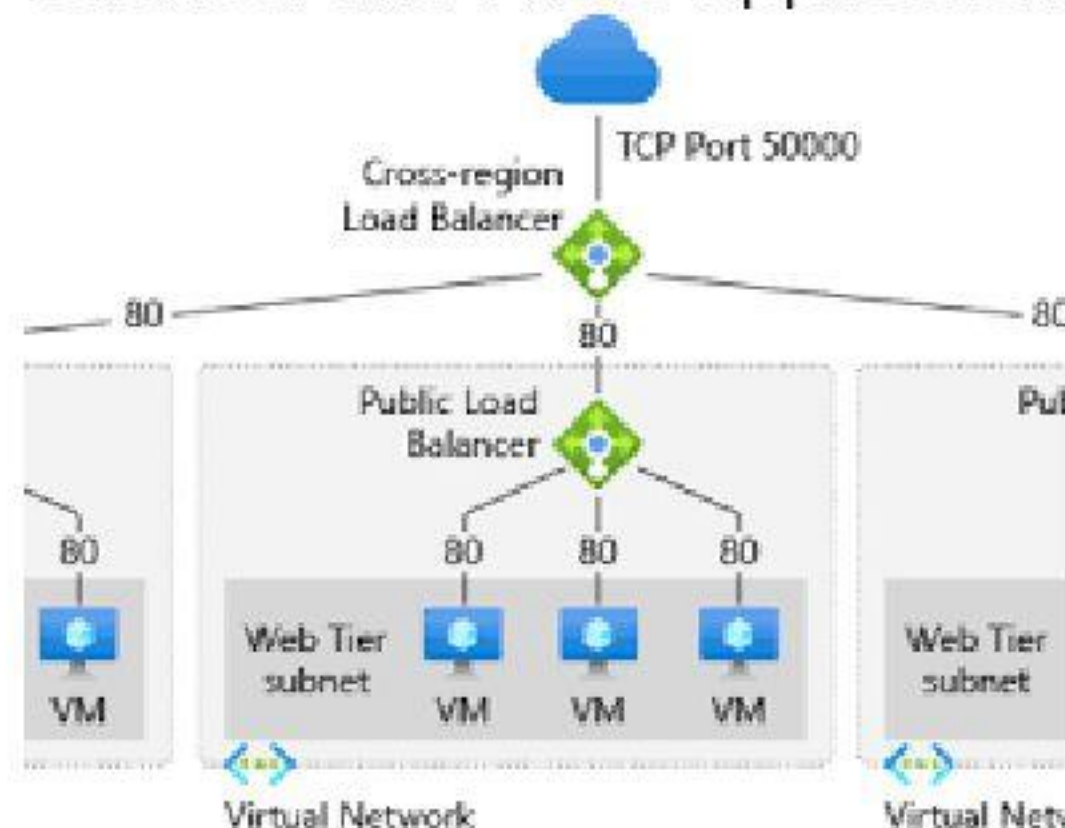
Microsoft Azure Serverless

Computing Networking

- Virtual Networks (VNet): A VNet is a private network in the cloud. It allows you to connect your VMs together and to the internet.
- Firewalls: Firewalls help to protect your cloud-based resources from unauthorized access. Azure offers a variety of firewall services, such as Azure Firewall and Azure Network Security Groups (NSGs).



Load Balancers: Load balancers distribute traffic across multiple VMs. This can help to improve the performance and availability of your applications. Azure offers a variety of load balancer services, such as Azure Load Balancer and Azure Application Gateway.



Storage

- Blobs: Blobs are a type of storage that is used to store unstructured data, such as images, videos, and documents. Azure offers a variety of blob storage services, such as Azure Blob Storage and Azure Data Lake Storage.

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services



Microsoft Azure Blobs

- Queues: Queues are a type of storage that is used to store messages. Messages are stored in the queue until they are processed by a worker role. Azure offers a variety of queue storage services, such as Azure Queue Storage and Azure Service Bus.

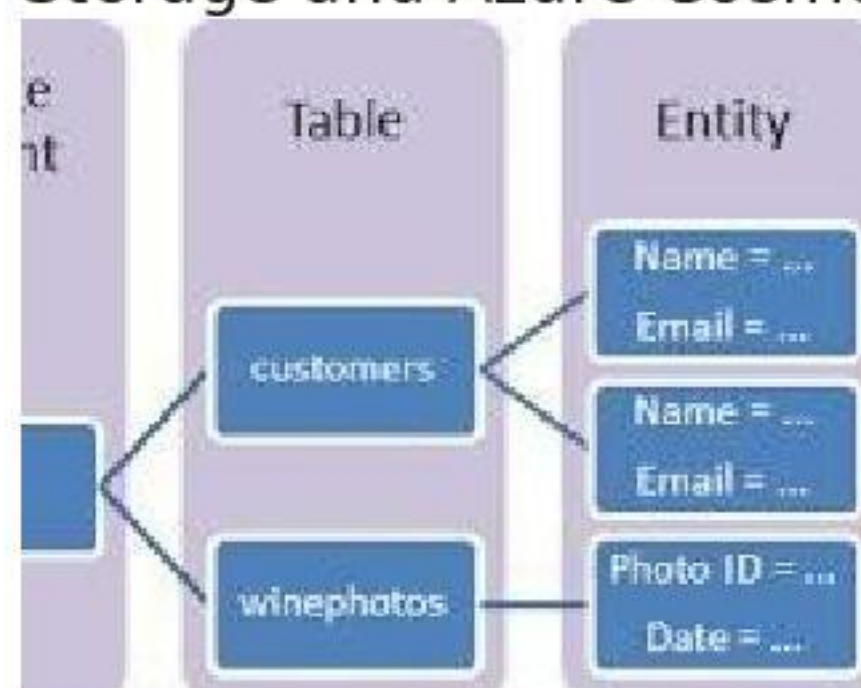
What is Azure Storage Queue?



Azure Storage Queue

Microsoft Azure Queues

- Tables: Tables are a type of storage that is used to store structured data. Tables are similar to databases, but they are simpler and more scalable. Azure offers a variety of table storage services, such as Azure Table Storage and Azure Cosmos DB.



Microsoft Azure Tables

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

Databases

- SQL Database: SQL Database is a fully managed relational database service that supports SQL Server workloads.



Microsoft Azure SQL Database

- Cosmos DB: Cosmos DB is a NoSQL database service that is designed for global scale and high performance. It supports a variety of APIs, including SQL, MongoDB, Cassandra, and Gremlin.



Microsoft Azure Cosmos DB

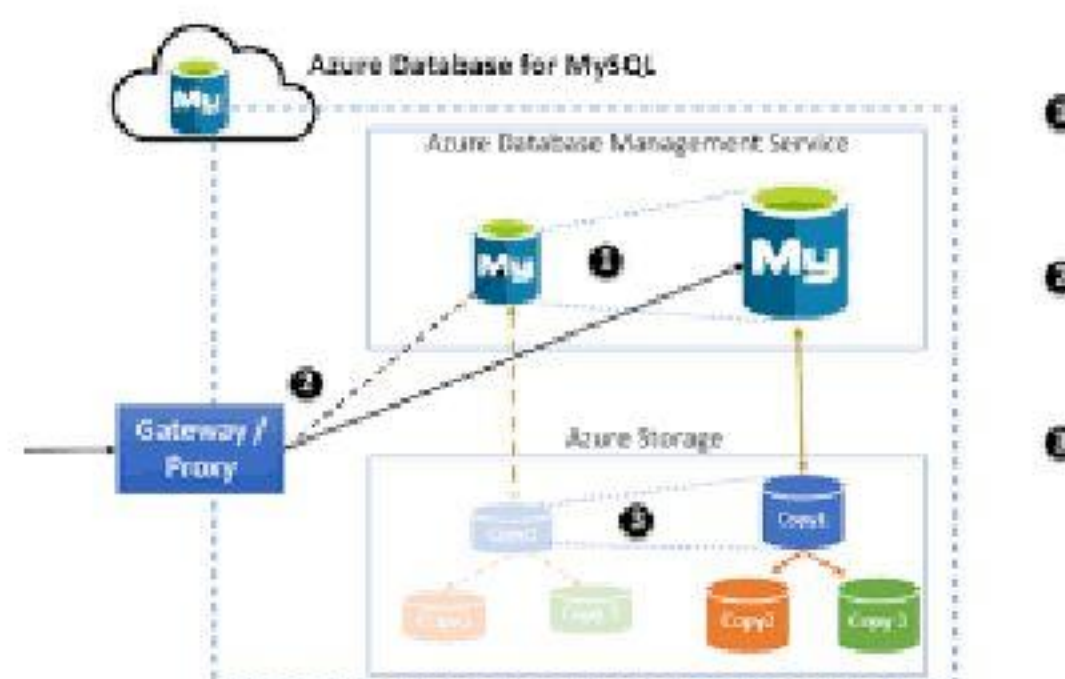
- MySQL Database: MySQL Database is a fully managed relational database service that supports MySQL workloads.

CLOUD COMPUTING-II

UNIT-III

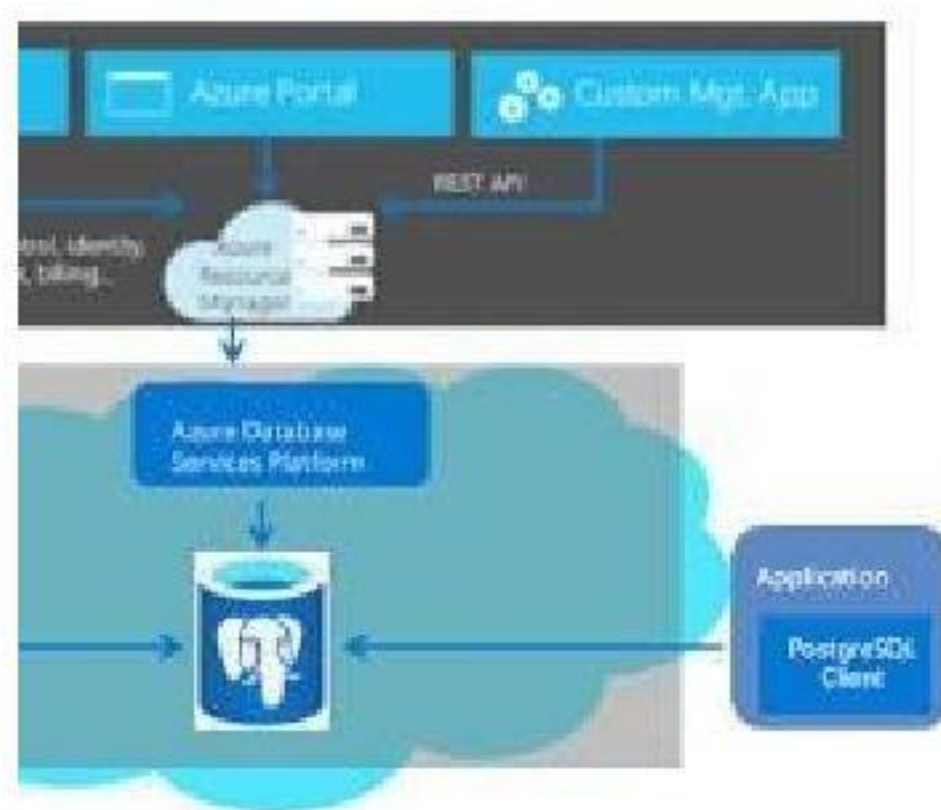
Microsoft Cloud Services

g with Built-in High Availability



Microsoft Azure MySQL Database

- PostgreSQL Database: PostgreSQL Database is a fully managed relational database service that supports PostgreSQL workloads.



Microsoft Azure PostgreSQL Database

These are just a few of the many core Azure services that are available. For more information, please visit the Azure website.

Management tools on Azure

Azure offers various tools to help you manage resources in your Azure subscription. These tools can help you to:

- Monitor and troubleshoot resource performance and health
- Configure and manage resource settings
- Automate tasks and workflows
- Enforce policies and compliance requirements
- Optimize resource utilization and costs

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

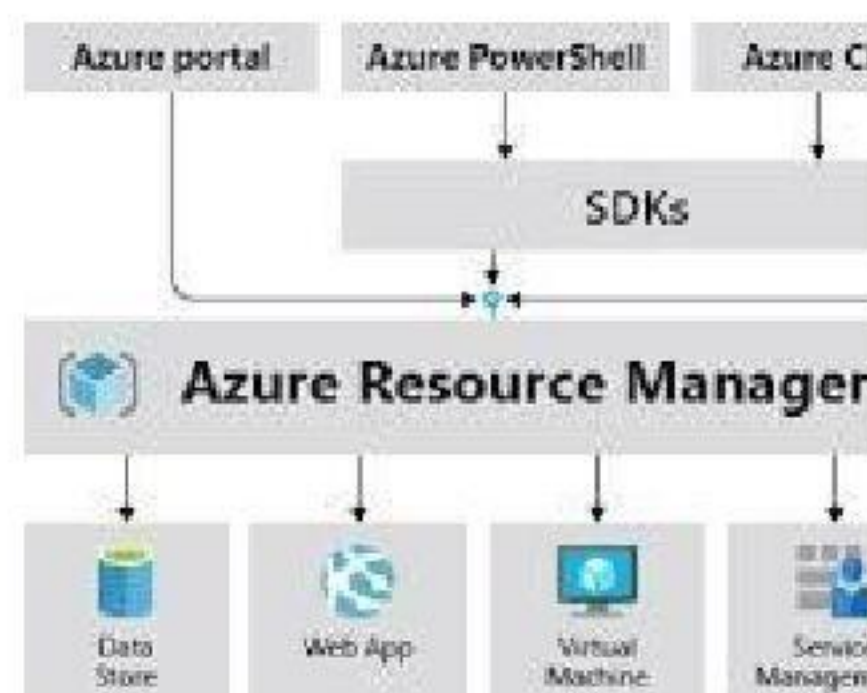
Here are some of the most popular Azure management tools:

- **Azure Monitor:** Azure Monitor is a comprehensive monitoring solution that provides insights into the performance and health of your Azure resources. It collects data from a variety of sources, including logs, metrics, and events. You can use Azure Monitor to diagnose problems, identify trends, and optimize resource utilization.



Microsoft Azure Monitor

- **Azure Resource Manager (ARM):** ARM is a cloud-based service that provides a centralized way to manage your Azure resources. You can use ARM to create, deploy, and manage your resources using templates or scripts. You can also use ARM to apply policies and enforce compliance requirements.



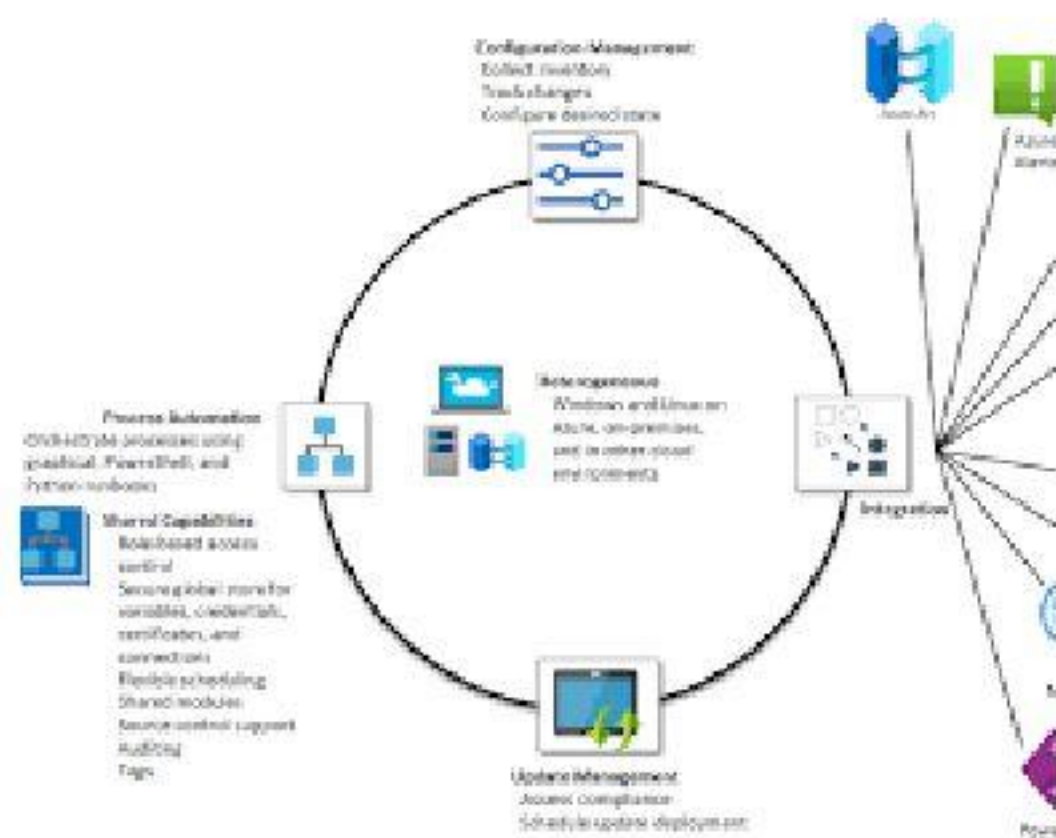
Microsoft Azure Resource Manager

- **Azure Automation:** Azure Automation is a cloud-based service that helps you automate tasks and workflows. You can use Azure Automation to create runbooks, which are scripts that can be used to automate tasks such as deploying software, managing configurations, and responding to events.

CLOUD COMPUTING-II

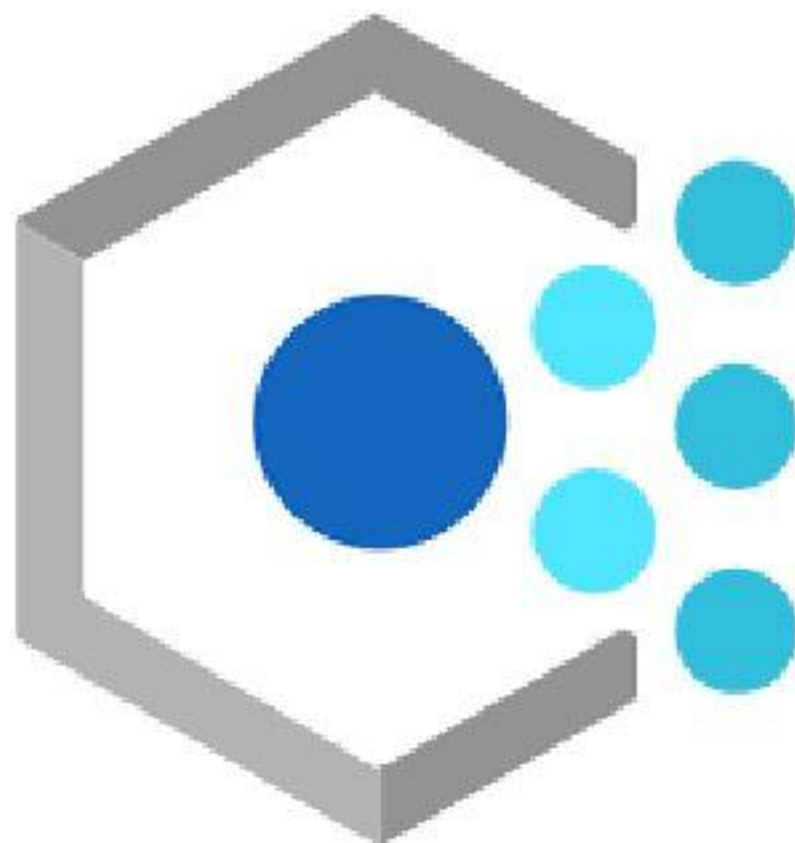
UNIT-III

Microsoft Cloud Services



Microsoft Azure Automation

- **Azure Policy:** Azure Policy is a cloud-based service that helps you enforce policies and compliance requirements. You can use Azure Policy to create policies that define rules for how your resources can be used. Azure Policy can automatically apply these policies to new resources and remediate existing resources that are not compliant.



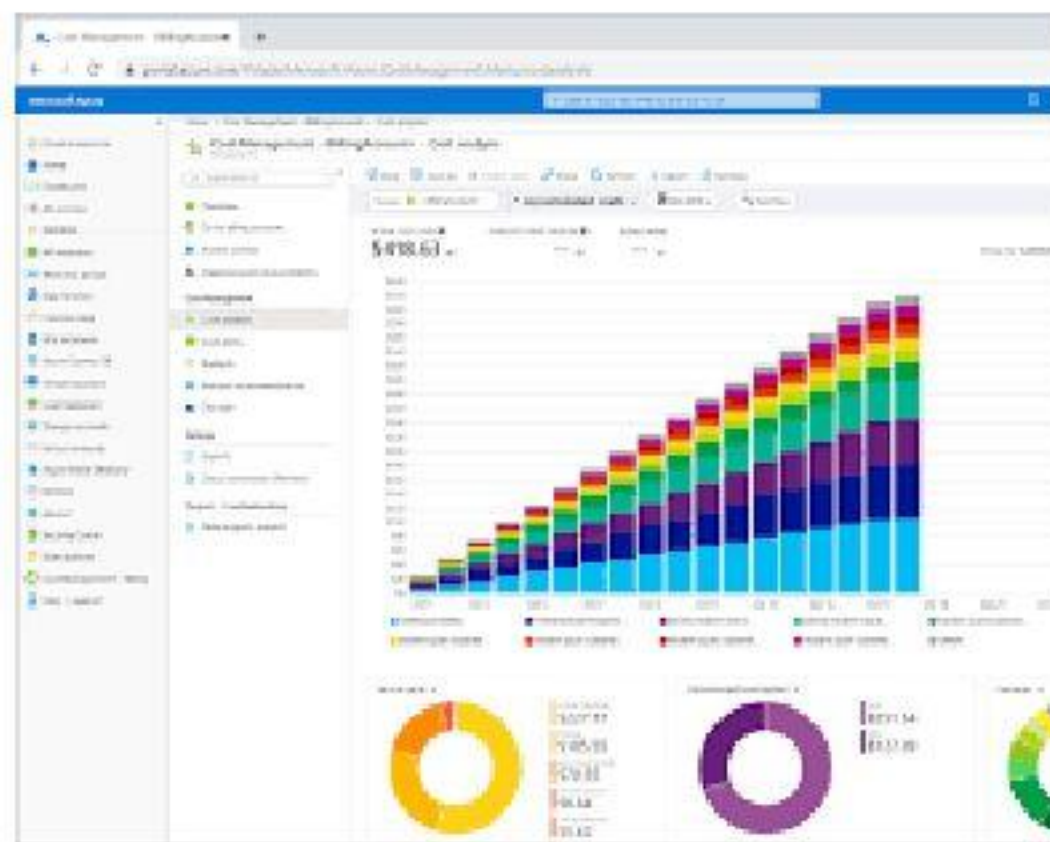
Microsoft Azure Policy

- **Azure Cost Management:** Azure Cost Management is a cloud-based service that helps you track and optimize your Azure spending. You can use Azure Cost Management to analyze your spending trends, identify areas where you can save money, and set budgets and alerts.

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services



Microsoft Azure Cost Management

In addition to these general-purpose management tools, Azure also offers a variety of tools for specific types of resources, such as virtual machines, storage accounts, and databases.

Security features of Azure

Azure offers a comprehensive suite of security features to help you protect your cloud resources, data, and applications. These features are designed to help you prevent, detect, and respond to security threats, and to comply with security regulations.

Key security features of Azure include:

1. Identity and access management (IAM): IAM helps you control who can access your Azure resources and what they can do with those resources. Azure IAM provides a variety of features, such as role-based access control (RBAC), multi-factor authentication (MFA), and just-in-time access (JIT).
2. Threat protection: Azure offers a variety of threat protection services, such as Azure Sentinel, Azure Defender for Cloud, and Azure Security Center. These services can help you detect and respond to threats, such as malware, ransomware, and phishing attacks.
3. Data protection: Azure offers a variety of data protection features, such as encryption, key management, and data loss prevention (DLP). These features can help you protect your data from unauthorized access, loss, or disclosure.
4. Network security: Azure offers a variety of network security features, such as firewalls, intrusion detection/prevention systems (IDS/IPS), and network

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

segmentation. These features can help you protect your networks from unauthorized access and malicious traffic.

5. Compliance: Azure offers a variety of compliance features, such as Azure Security Center for Compliance and Azure Compliance Manager. These features can help you assess and comply with security regulations, such as HIPAA, PCI DSS, and SOC 2.
6. Security monitoring and alerting: Azure offers a variety of security monitoring and alerting features, such as Azure Monitor, Azure Log Analytics, and Azure Sentinel. These features can help you identify and investigate security incidents in real time.
7. Security automation: Azure offers a variety of security automation features, such as Azure Automation and Azure Logic Apps. These features can help you automate security tasks, such as vulnerability management, incident response, and compliance enforcement.
8. Security training and resources: Microsoft provides a variety of security training and resources to help you learn about Azure security and how to protect your cloud resources. These resources include documentation, videos, and training courses.

By leveraging these security features, you can create a secure cloud environment that protects your data, applications, and infrastructure from unauthorized access, theft, and damage.

Introduction to Azure SQL

Azure SQL is a family of fully managed, intelligent, and scalable cloud database services that builds on the rich history of Microsoft SQL Server. It provides a wide range of features and capabilities that enable you to build, deploy, and manage highly scalable and performant relational databases in the cloud.

Key features of Azure SQL include:

- Fully managed: Azure SQL is a fully managed service, which means that Microsoft takes care of all the infrastructure and database administration tasks, such as provisioning, patching, and backups. This frees you up to focus on developing and managing your applications.
- Intelligent: Azure SQL provides a number of intelligent features that can help you optimize your database performance and cost. These features include Intelligent

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

Insights, which provides recommendations for improving your database performance, and Adaptive Query Acceleration, which automatically optimizes your queries.

- ↯ Scalable: Azure SQL can be scaled up or down to meet your changing needs. You can easily add or remove compute and storage resources to your database without having to worry about provisioning or managing infrastructure.
- ↯ High performance: Azure SQL is designed for high performance and can handle even the most demanding workloads. It offers a variety of service tiers, each with different performance and cost characteristics.
- ↯ Secure: Azure SQL is a highly secure cloud database service that is compliant with a wide range of industry standards, including HIPAA, PCI DSS, and SOC 2.

Benefits of using Azure SQL:

There are many benefits to using Azure SQL, including:

- ↯ Reduced costs: Azure SQL can help you reduce your costs by eliminating the need to purchase and maintain on-premises hardware and software. You only pay for the resources you use, and you can scale up or down as needed.
- ↯ Increased agility: Azure SQL can help you increase agility by allowing you to quickly provision and scale your database resources. This can help you respond to changes in demand and take advantage of new opportunities.
- ↯ Improved productivity: Azure SQL can help you improve productivity by providing your developers with access to a highly performant and reliable database service. This can free them up to focus on developing and deploying applications.
- ↯ Enhanced security: Azure SQL is a highly secure cloud database service that is compliant with a wide range of industry standards. This can help you protect your data from unauthorized access, theft, and damage.
- ↯ Empowered innovation: Azure SQL can help you empower innovation by providing you with the tools and resources you need to develop new products and services.

Use cases of Azure SQL:

Azure SQL can be used for a variety of use cases, including:

- ↯ Web applications: Azure SQL can be used to store and manage data for web applications.

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

- ↗ Mobile applications: Azure SQL can be used to store and manage data for mobile applications.
- ↗ Internet of Things (IoT): Azure SQL can be used to store and manage data from IoT devices.
- ↗ Big data: Azure SQL can be used to store and analyze big data.
- ↗ Machine learning: Azure SQL can be used to store and train machine learning models.

Overall, Azure SQL is a powerful and versatile cloud database service that can be used to address a wide range of business needs.

Here are some additional resources that you may find helpful:

- ↗ Azure SQL website: <https://azure.microsoft.com/en-us/products/azure-sql>
- ↗ Azure SQL documentation: <https://learn.microsoft.com/en-us/azure/azure-sql/?view=azuresql>
- ↗ Azure SQL blog: <https://techcommunity.microsoft.com/t5/azure-sql-blog/bg-p/AzureSQLBlog>

Hosting and provisioning with Azure

refers to the process of creating, managing, and deploying resources in the Microsoft Azure cloud platform. It involves a range of tasks, from setting up virtual machines and containers to configuring networking and storage. Azure's hosting and provisioning capabilities enable businesses to build and maintain scalable, reliable, and secure cloud-based infrastructure.

Key Concepts in Hosting and Provisioning with Azure

Resource Groups: Resource groups are logical containers that help you organize and manage your Azure resources. They allow you to group related resources together, apply policies and tags, and simplify access control.

Virtual Machines (VMs): VMs are fundamental building blocks of cloud computing that provide a dedicated environment for running applications. Azure offers a variety of VM options, including Windows Server, Linux, and specialized VMs for specific workloads.

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

Containers: Containers are lightweight, portable units of software that package code, dependencies, and runtime environments. Azure provides various container services, such as Azure Container Instances (ACI) and Azure Kubernetes Service (AKS), to manage and orchestrate containerized applications.

Storage Accounts: Storage accounts provide durable and scalable storage for various data types, including blobs, queues, and tables. Azure offers different storage tiers, such as Standard and Premium SSD, to meet performance and cost requirements.

Networking: Networking in Azure involves configuring virtual networks, firewalls, and load balancers to connect and secure cloud resources. Azure's networking services enable efficient traffic routing and protection from unauthorized access.

Provisioning Resources in Azure

Azure offers various methods for provisioning resources, including the Azure portal, the Azure CLI, and Azure PowerShell. These tools allow you to create, configure, and manage resources through a user interface, command-line interface, or scripting language, respectively.

Provisioning with Templates: Azure Resource Manager (ARM) templates provide a declarative way to define and provision infrastructure resources consistently and repeatably. Templates use JSON or YAML syntax to describe the desired state of resources, such as VMs, storage accounts, and networking components.

Benefits of Hosting and Provisioning with Azure

Scalability: Azure's elastic infrastructure allows you to scale resources up or down seamlessly to meet changing demands.

Cost-effectiveness: Azure's pay-as-you-go model allows you to only pay for the resources you use, optimizing costs based on usage patterns.

Reliability: Azure's global data centers and redundancy features ensure high availability and disaster recovery for your cloud applications.

CLOUD COMPUTING-II

UNIT-III

Microsoft Cloud Services

Security: Azure provides a comprehensive suite of security features, including identity and access management, threat detection, and data encryption, to protect your cloud infrastructure and data.

Agility: Azure's self-service provisioning tools and automation capabilities enable rapid deployment and management of cloud resources.

Overall, hosting and provisioning with Azure provides a flexible, cost-effective, and secure platform for building, deploying, and managing cloud-based applications and infrastructure.